

№8

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0.1



$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f(t)$
 $(y = \dot{x})$

0.2



:

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = f(t)$$



$(y = \dot{x})$:

$$\begin{cases} \frac{dx}{dt} = y \\ \frac{dy}{dt} = -2\gamma y - \omega_0^2 x + f(t) \end{cases}$$

0.3

Julia

“julia using DifferentialEquations using Plots

w0_sq = 3.5 gamma = 0.5 u0 = [1.0, -0.5] tspan = (0.0, 50.0)

1. 1:

```
function model1!(du, u, p, t) du[1] = u[2] du[2] = -w0_sq * u[1]
end sol1 = solve(ODEProblem(model1!, u0, tspan), saveat=0.05)
```

2. 2:

```
function model2!(du, u, p, t) du[1] = u[2] du[2] = -2 * gamma *  
u[2] - w0_sq * u[1] end sol2 = solve(ODEProblem(model2!, u0,  
tspan), saveat=0.05)
```

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3. 3:

```
f(t) = 0.7 * sin(2.0 * t) function model3!(du, u, p, t) du[1] = u[2]  
du[2] = -2 * gamma * u[2] - w0_sq * u[1] + f(t) end sol3 =  
solve(ODEProblem(model3!, u0, tspan), saveat=0.05)
```

4. —

grafik 1(?@fig-001)

```
[59]: # Stasyoner noktaların cebirsel denklemlerden çözümü (1c denge durumu  $x_2 > 0$  ve  $x_2 > 0$ )
# c1 =  $a_1^2 x_1 - b^2 x_2 = 0$ 
# c2 =  $a_2^2 x_2 - b^2 x_1 = 0$ 

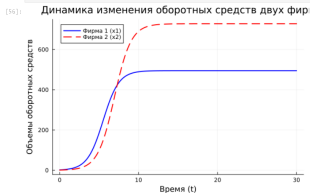
# Matris çözümleri (d = X = 0)
A = [a1 b; b a2]
B = [c1, c2]
steady_state = A \ B

println("---- Stasyoner nokta durumu ----")
println("Stasyoner nokta durumu formu 1 (x1*): ", steady_state[1])
println("Stasyoner nokta durumu formu 2 (x2*): ", steady_state[2])

---- Stasyoner nokta durumu ----
Stasyoner nokta durumu formu 1 (x1*): 494.5415691568835
Stasyoner nokta durumu formu 2 (x2*): 727.5381569156883
```

[]:

grafik 2(?@fig-002)



#

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2-

1-

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